Reference Paper A - Algebra, Number and Ratio (O Level style)

MARK SCHEME

Lesson: Ch2L3
Total: 29 marks
Marks in brackets are the marks available for that part.

Question 1 [3 marks]
Solve 4x + 7 = 31.
M1: 4x = 24 seen (subtract 7 from both sides)
M1: divide both sides by 4
A1: x = 6 (cao)

Question 2 [2 marks]
Write 0.36 as a fraction in its simplest form.
M1: 36/100 seen
A1: 9/25 (cao)

Question 3 [3 marks]
The nth term of a sequence is 3n - 2. (a) Write down the 4th term. (b) Which term of the sequence equals 40?
B1: 4th term = 10
M1: sets 3n - 2 = 40
A1: n = 14 (cao)

Question 4 [2 marks]
Work out (2^3)^2.
M1: 2^6 seen (or 8 squared)
A1: 64 (cao)

Question 5 [2 marks]
Expand (x + 3)(x - 5).
M1: four correct terms or a complete correct area/grid product (x^2, -5x, 3x, -15)
A1: x^2 - 2x - 15 (cao)

Question 6 [3 marks]
Simplify 12a^5 b^3 / (4a^2 b).
M1: coefficient 3
M1: a^3
A1: 3a^3 b^2 (or any equivalent ordering such as 3b^2 a^3)

Question 7 [2 marks]
Factorise x^2 + 7x + 12.
M1: (x + 3)(x + 4) seen
A1: both factors correct (cao)

Question 8 [3 marks]
Solve simultaneously: 2x + y = 11 and x - y = 1.
M1: adds (or otherwise eliminates) to get 3x = 12
M1: x = 4

A1: y = 3 (cao, dependent on a correct x)

Question 9 [2 marks]

A triangle has base 12 cm and perpendicular height 9 cm. Work out its area.

M1: 1/2 x 12 x 9 (correct substitution)

A1: 54 cm^2 (cao, units required)

Question 10 [2 marks]

Hence solve x^2 + 7x + 12 = 0.

M1 (ft): solves their own factorisation from the previous part

A1 (ft): the two roots consistent with that factorisation

Question 11 [2 marks]

Solve 5x - 3 = 2x + 9.

M1: 3x = 12 seen

A1: x = 4 (cao, a single unambiguous value required)

Question 12 [3 marks]

A car travels 150 km in 2.5 hours. Work out its average speed.

M1: 150 / 2.5 seen

A1: 60 (cao)

B1: km/h stated